

Laborator 2

Configurarea VLAN pe SWITCH

Scopul acestui laborator:

- Crearea VLAN-urilor pe un switch
- Asocierea porturilor la VLAN-uri
- Verificarea separării traficului între VLAN-uri

1. Crearea topologiei:

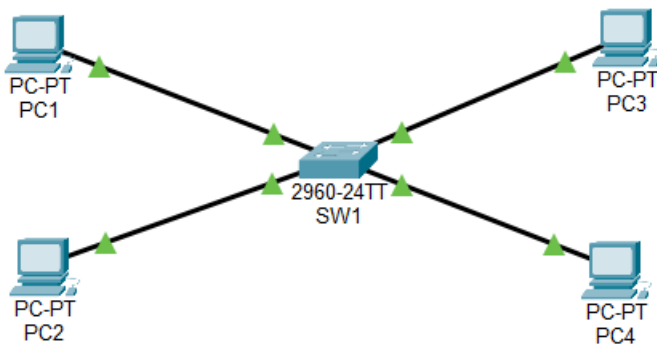
Adăugați următoarele dispozitive:

- 1 x Switch (ex: **2960**)
- 4 x PC

Conectați dispozitivele astfel:

Device		SwitchPort
PC1	->	Fa0/1
PC2	->	Fa0/2
PC3	->	Fa0/3
PC4	->	Fa0/4

Topologie:



2. Configurarea adreselor IP pe PC-uri

Pe fiecare PC:

Desktop → IP Configuration

VLAN	PC	IP
10	PC1	192.168.10.2/24
10	PC2	192.168.10.3/24
20	PC3	192.168.20.2/24
20	PC4	192.168.20.2/24

3. Accesarea CLI pe switch

Click pe Switch → CLI

Inițializați modul privilegiat: **enable**

Intrarea în modul de configurare: **configure terminal**

4. Crearea VLAN-urilor

Vom crea două VLAN-uri:

- VLAN 10 - IT
- VLAN 20 – HR

```
Switch(config)#vlan 10
Switch(config-vlan)#name IT
```

Verificam crearea VLAN-urilor:

```
Switch#show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/1, Fa0/2, Fa0/3, Fa0/4 Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	IT	active	
20	HR	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

5. Configurarea porturilor pentru VLAN-uri

PC1 și PC2 vor face parte din VLAN 10.

Pentru PC2, fa0/2.

```
Switch(config)#interface fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 10
Switch(config-if)#no shutdown
```

PC3 și PC4 vor face parte din VLAN 20.

Verificare:

```
Switch(config)#do show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/5, Fa0/6, Fa0/7, Fa0/8 Fa0/9, Fa0/10, Fa0/11, Fa0/12 Fa0/13, Fa0/14, Fa0/15, Fa0/16 Fa0/17, Fa0/18, Fa0/19, Fa0/20 Fa0/21, Fa0/22, Fa0/23, Fa0/24 Gig0/1, Gig0/2
10	IT	active	Fa0/1, Fa0/2
20	HR	active	Fa0/3, Fa0/4
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

6. Testarea conectivității

```
C:\>ping 192.168.10.3
```

```
Pinging 192.168.10.3 with 32 bytes of data:
```

```
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
```

```
Reply from 192.168.10.3: bytes=32 time<1ms TTL=128
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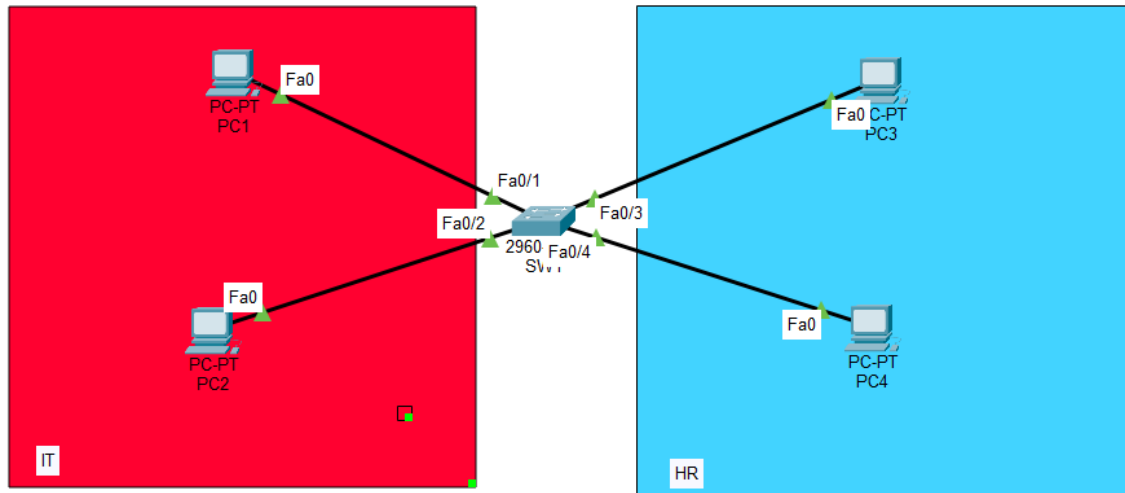
```
Ping statistics for 192.168.10.3:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),  
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

7. Salvarea configurației

copy running-config startup-config



8. Incarcati fisierul pe CV.